Reasons for Persons, or The Good Successor Problem

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November 24, 2024

Abstract

Short version

This doctoral project considers humanity's potential for evolution into a technologically mature species, and the values that ought to be transmitted to our successors. The written component reviews the historical context and current concerns of "AI alignment"; examines the variety of possible cognitions as a way of decentring anthropocentric biases; analyzes subject-neutral axiology (the "view from nowhere") and its criticisms; and considers how acausal decision theories might help evolve conceptions of morality that better suit an AI-dominated world.

The practice component involves collaborating with AI alignment researchers and using large language models (LLMs) as research partners and experimental subjects. This includes:

1) Analyzing how LLMs approach Newcomb's Paradox to gain insight into how agents with strong mutual predictive abilities might reason about each other's actions
2) Using LLM reasoning on acausal decision theory to inform how likely advanced AI may be to behave benevolently towards humans
3) Testing if LLMs incorporated into agentic structures actually behave as predicted in simu-

lated environments

The project aims to contribute new knowledge by providing a written history of AI alignment, integrating acausal decision theories with views-from-nowhere, and conducting novel practical investigations into LLM cognition and behavior in the context of AI alignment and safety. The overall goal is to (try to) deconfuse medium/long-term issues in AI development and hopefully improve the chances of a positive long-term future for Earth-originating intelligence.

Long version

Can humans evolve into a technologically mature species[1] in a way that is (in some sense) "safe"? How might current humans think about the "shape" and "values" of our (potentially, radically different) successor species?

The written element of the project has four components:

- A review of the historical origins, and current live concerns, of "AI alignment".[2] The worries about extinction-level risks exemplified in the AI alignment discourse are alleged (often pejoratively) to stem from, or be a variety of, historical preoccupations with eschatology, salvation, and human essentialism found in Abrahamic faiths. I examine this lineage, particularly in the Early Church and Calvinist/Puritan thought, but also suggest a more productive adjacency with the conception of markets (in Nick Land's reading of Hayek via Deleuze-Guattari and Bataille) as a species of "artificial intelligence" (understood as distributed computation)

[1] "Technologically mature" means a species that has shed the constraints of biology (though some or all members might still choose to maintain a biological form, for whatever reason), more fully utilises the endowment of low-entropy energy, and is capable of spreading artifacts to the stars.

[2] "AI alignment" inherits from theories of how rational agents make decisions under uncertainty, which combined with transhumanist/extropian futurism that was current in the 1990s-2000s.

which, through a type of impersonal teleological agency, are giving birth to (silicon-based) AI.[3]

- A review of existing writing on the different ways cognition can manifest, with respect to the constraints of physics, as well as existing examples in biological systems. The possibility of radically different organisations-of-mind is historically foundational to the AI alignment discourse. It also sets the stage for the project's overall argument: that humans and human values are contingent, and only "worth preserving" insofar as they are invariant to any particular biological and cultural constitution and history. That is, my claim is that the values that another species holds could be otherwise (and conversely, those we hold dear are mostly arbitrary), except to the extent that they are grounded in (an adequately enriched and robust notion of) rationality.
- A historical review of the subject-neutral perspective, that sometimes goes under terms like "point-of-view of the universe" (Henry Sidgwick/Peter Singer/G.E. Moore), "view from nowhere" (Thomas Nagel), or Derek Parfit's "triple theory". I also examine some reasons such views are rejected (e.g. Bernard Williams). My provisional claim is that these ideas (and their criticisms) might be worth revisiting if, in fact, humanity is on the cusp of creating new forms of intelligence, which might not always reason about values or the world under the same constraints and foundational assumptions (e.g. they aren't conditioned to the same degree by human biological and cultural evolution). If we refuse to consider a less parochial perspective, how will we justify our actions (morally) in respect of current or future synthetic intelligences? Conversely, to the extent that we must convince our AI successors to behave

[3] Land's writing in this vein dates from the CCRU period of the 1990s and has been revisited in 2023/2024; the parts of interest are fairly distinct from his neo-reactionary views.

beneficently towards life on Earth (or all sentient beings), we must provide reasons that are as universal as possible.

- One criticism of the universal point-of-view is that we cannot reason about something that is out of our distribution[4], which (I claim) can be seen as a weak form of the incompleteness/incomputability originally suggested by Gödel, Turing, Church, as well as others. This, in turn, is intimately related to the problem of alignment, particularly of super-human AI.[5] One (imperfect and provisional) approach is the use of acausal decision theories, which adapt traditional game/decision theory to the constraints and affordances advanced AI systems might face. More speculatively, these theories are extended in the literature to "large worlds" (e.g. astronomical scales with aliens, some of which might be rational agents), where they abut David Deutsch's Popperian proposal of morality as a thing that is continually evolving, creatively and discursively. Deutsch's dynamic morality, in turn, recalls Land's invocation of the Confucian idea of "self-cultivation" as the hallmark of intelligence, as well as Reza Negarestani's Platonic framing of intelligence as a realisation of the Good, something that arises communally through the computational medium of language.

The practice element of this project involves collaboration with AI alignment researchers. Hence actual research tasks might change, but these are a few current investigations:

- Large language models (LLMs) act as (my) research partners as well as experimental subjects. Specifically, I consider Newcomb's Paradox in detail, the findings of which inform the

[4]I use "distribution", a term from AI research, to gesture at humanity's collective body of experience (as documented) as well as the basic constraints imposed by our individual and collective intellectual capacity.

[5]The relation comes from questions like: "How does a rational agent reason about systems within which it is embedded?", "How does a rational agent shape the goals and decision-making of a more powerful agent?", "How does a rational agent ensure the goals and decision-making structure it has instilled in another thinking system persist when that (second) system creates a (third) thinking system?".

discussion on alignment. [6] This investigation is carried out through behavioural analysis (akin to test questions in human psychology) and neural probes (directly accessing the LLM's internals).[7]

- LLM reasoning on acausal decision theory, in turn, might inform researchers on how likely advanced AIs are to be beneficently-disposed towards humans (which, after all, is the core aim of AI alignment, safety, and ethics). This latter question is speculatively posed through an "epistle to the successor" (a letter to a future superintelligence asking it to behave benevolently towards humans), and then analysed through debate amongst LLMs, in an instantiation of a "moral parliament" (Toby Ord).
- LLMs are often framed as "simulators" of "personas".[8] Hence, their answers to questions might have little bearing on what they actually do in the world. One way of testing this is by incorporating LLMs within structures (known as "LLM agents") that allow them to take actions in simulated or real-world situations. This agentic scaffolding approach is used to test whether current models actually behave as one-boxers (with respect to Newcomb's Paradox) in a simulated environment.

The contributions to knowledge are:

- At present, there doesn't seem to be a written history of AI alignment, or a well-considered

[6]Newcomb's Paradox is a puzzle in acausal decision theory first analysed by Robert Nozick, which might offer insight on how agents with strong mutual predictive abilities should reason about each others' actions; it is a situation found in human affairs, but might become more the case in multi-AI environments.

[7]Neural probes raise nascent questions around suffering risks, in respect of (possibly) conscious AI systems. These are important, but probably not a risk on current LLMs. Absent a change in consensus, the ethics around AI suffering are treated as out-of-scope for this project.

[8]The "simulators" framing was suggested in 2022 by the collective Janus, and is a newer, more sophisticated presentation of LLM cognition than "stochastic parrots" (Emily Bender and others). Simulators' idea is that LLMs will say almost anything (within the constraints of what they were trained upon) in response to a prompt, and that they don't have any true "beliefs" about the world or themselves.

analysis of adjacencies with Landian accelerationism or Christian eschatology.

- Discussions around decision theory for super-rational agents currently sits in an online, rationalist "bubble" or in narrow academic niches, and doesn't seem to have been integrated with (or criticised with reference to) views-from-nowhere.
- All practical investigation of how LLMs approach these questions is, by definition (owing to the rapid evolution in AI capabilities), new.